

Kinect for katas

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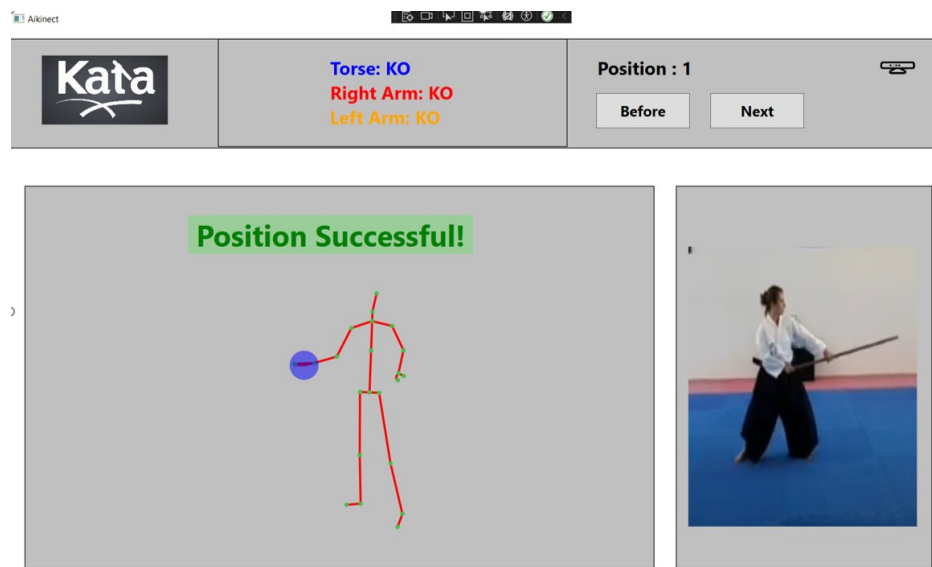
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If you have any problems recognising the skeleton, please remove any loose clothing.

ABSTRACT

In our work, we developed an angle-based position validation system for kata in aikido martial art where kata refers to a sequence of gestures and movements in traditional Japanese martial arts. Upper-body bone angles of pre-defined kata are registered in a list. The system use the Kinect sensors to estimate the coordinates of the body joints and then calculate the angle of the bones with them. If our bone angles correspond to the current tested position then the position is validated.

The result of our work proves to be difficult with loose clothes (used in aikido) and some position are hard to recognise.

CCS CONCEPTS

• **Human-centered computing** → **Mixed / augmented reality**; Graphical user interfaces; User interface programming; *Interaction devices*.

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1 INTRODUCTION

Using Microsoft's Kinect for Aikido position recognition can push the system to its limits due to several factors that will be outlined later. We wish to test the limitations of the Kinect when using a simple recognition system rather than more advanced Machine Learning algorithms.

Correcting kata position at home proved to be difficult due to the lack of any large mirror or coach to correct our posture. This program could resolve this problem by providing help to users by validating correct positions. Moreover, coaches could easily add

new position to recognise simply by adding new bones angles. The lack of learning algorithms makes adding new positions to the program much easier and quicker.

The concept of 'kata' is essential in many martial disciplines, including Aikido, where the precision of movements and body positions is crucial. A kata refers to a pre-established sequence of movements and techniques that are practised in a ritualistic manner. The algorithm could be used to help practitioners perfect their kata by providing instant feedback on the precision of each movement, thus contributing to the overall improvement of their martial performance.

2 RELATED WORK

This section contains a list of 4 research articles aimed at motion capture with the use of the Kinect. The articles found and retained after analysis are as follows:

1st article: Recognition of Yoga Poses Through an Interactive System with Kinect Device [4].

Description: Gesture recognition of 6 Yoga poses for up to 6 users using the Kinect and the AdaBoost algorithm used to train a database. Also using voice commands.

2nd article: New Design System to Learning Martial Art Via Kinect 2.0 [3].

Description: Learning Thai Boxing via the Kinect. Accompanied by an RGBD sensor connecting the user to a professional trainer, who recovers the apprentice's movements in real time.

3rd article: In search of a usability of Kinect in the training of traditional Japanese "KATA" — Stylized gestures and movements [5].

Description: This study explores the use of Kinect for learning Kata.

4th article: Virtual Kung fu Sifu with Kinect [1].

Description : Master's project focusing on Kung Fu, the movements are pre-recorded with the Kinect, they will be compared with the user's movements using the Dynamic Time Wrapping algorithm using Euclidean distance. article5

5th article: Gesture Recognition from Indian Classical Dance Using Kinect Sensor [2].

Description : Recognition of type of dance based on angle, velocity and acceleration.

3 METHODOLOGY

3.1 Position Recognition Program

To recognize positions, we decided to use values native to the Kinect. Indeed, the Kinect natively returns the position of each joint of the body. To avoid dealing with people of differing limb sizes we decided to use the slope of the limbs of the user. Since we already had all the joints, we can consider a limb to be the link between two joints and the slope to be $\frac{\Delta x}{\Delta y}$. The problem with using simple slopes is that they don't increase linearly, the difference between a slope of 0 and 1 is much bigger visually than the difference between a slope of 50 and 100. To solve that issue, we decided to instead use

the angle created by the limb. To obtain that angle, we can refer to trigonometry and use the formula $\frac{\Delta y}{\Delta x} = \tan \theta$ and we can obtain θ using $\theta = \arctan \frac{\Delta y}{\Delta x}$. While using angles solves our linearity problem, we still have the issue that angles jump when they get to a certain value. Either we jump from 180 to -180 or we jump from 360 to 0. To circumvent that problem, we use angles that go from 0 to 180 degrees and without negatives by using absolute values. Once our values for each limb are registered, we simply the registered values with the values that the Kinect is giving to us with a certain delta to make the program more lenient.

3.2 Registering the bone angles

A brown belt aikidoka helped us register the upper-body bone angles of 9 position of the 31st kata. We record the different angles in an XML file with the Kinect sensors and write the data down to create our list of positions.

We compare this list with our bone angles and validate the position if the values correspond.

3.3 Problems encountered linked to the katas

During those steps we have encountered several different problems. Most of those problems were concerning the reading of the movements by the kinect. We had to do a few choices due to that. Indeed generally in Aikido, the aikidoka wears a hakama (a sort of really large pant) over a kimono, both hide a lot of subtle movements. To help the kinect detecting the movements we consider that the target of this product would be the aikidoka training at home for their upcoming exam or demonstration and so he would be in regular civilian clothes in his living room.

Another limitation that we couldn't resolve was with the detection of the different katas themselves. Indeed the katas aren't always facing the same directions and sometimes the movements are very close to the body which makes it hard for the kinect to detect properly. To help fix this problem we decided to put the kinect on the right side of the main direction of the aikidoka. There are still some movements the kinect struggles to detect properly but having the kinect on that side improve greatly the detection.

We also decided to only detect the upper-body as the kinect had a lot of problem detecting the lower-body and since the upper-body has less subtle movement and more drastic changes it is easier to detect furthermore it is more important when beginning to learn the katas.

3.4 Problems encountered with the position recognition program

Finding an appropriate delta for the comparison of angles has proven to be complicated. We have to find a value that prevents the program from being too lenient but also we don't want to have a program that is too strict for the positions where a user might spend several seconds correcting limbs that don't need any correction. Another problem with the program's leniency is the similarity of different positions. We have more than once been into situations

were a user would get into a position and the program would validate more than one position. This is most likely due to the fact that positions look strongly alike when a program is just using the angles of separate limbs.

4 CONTRIBUTION

This work provide a very simple method to recognize body position with the Kinect. However, very few articles use the bone angle to recognize body position or movement.

One of the few studies that use angles is *The Gesture Recognition from Indian Classical Dance Using Kinect Sensor* [2]. This study recognize very general type of movement (e.g. the *sadness* movement is when the angle is decreasing continuously) and do not recognize precise positions.

Unlike them, we want the Kinect to recognize an accurate position, similar to what an Aikido teacher might expect and that is why we faced entirely different problems.

Our method is probably the simplest way to make a program that recognizes these kinds of position but we do believe that the intricacies of Aikido makes it very inconsistent.

5 EVALUATION

We benefited from the presence of an aikidoka expert during our stance data recording sessions. The stances were performed several times by the expert to ensure the greatest possible accuracy. After performing the stances, we reproduced them with individuals familiar with martial arts, under the expert's supervision. The aim of this approach was to ensure mutual validation between the program and the expert for each position.

Sometimes people more unfamiliar with the kata had easier time detecting the correct position as they rely more on the application contrary to experts who are not used to stop that long when performing the movements.

6 DISCUSSION

Using angles to position limbs on a body that we can use to recognize martial arts position has its limitations but also its advantages. First, the program is very simple at its core and requires very little set-up. The method does fall short when it comes to several factors. The program is either too lenient or too strict and it has been very difficult to find a proper middle ground.

7 CONCLUSION

In conclusion, our work presents a new approach to position recognition and validation in kata, using Microsoft Kinect sensors as like the 'expert'. The development of an angle-based position validation system, focusing on the angles of the upper body bones, mainly the arms, in predefined kata, has been described. The methodology involves using Kinect's native values to calculate the slopes and angles of the limbs, enabling specific positions to be recognised.

Our results, however, reveal certain difficulties caused by the loose clothing that can be worn by users, leading to inaccuracies in position recognition. Also, some of the kata positions proved difficult to identify precisely. Apart from a few difficulties, our work

shows that it can help people who want to practise kata at home, by providing real-time feedback on each member.

OPEN SCIENCE

Our GitHub repository with code is accessible at https://github.com/LeigerMax/UNAMUR-INFOM435-Projet_Kinect_Kata.

A demonstration video is also visible at https://youtu.be/-e67oEA_8Y

ACKNOWLEDGMENTS

We're grateful to GitHub user violetasdev for writing an in-depth course on body tracking with the Kinect's depth cameras. The course is available at: https://github.com/violetasdev/bodytrackingdepth_course

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